DEEP LEARNING SUMMER SCHOOL 2026

Domain Adaptation

Building financial and climate intelligence

Name the failure you can measure, then choose the axis that removes it — and build the evaluation before the adapter.

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Where you are

A general model, a budget, and a domain that is not the internet

Start with a capable general model. Then ask what the provider cannot supply:

- your evidence and decision date;
- your label and economic estimand;
- your loss and deployment gate.

Given a deployment failure you can *measure*, where should the domain signal enter?

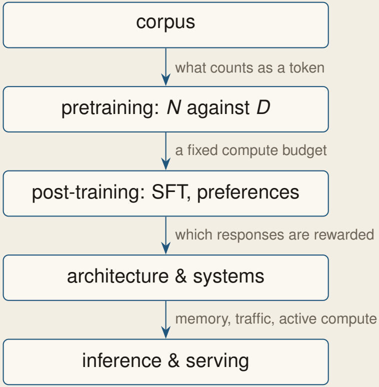
The discipline in one sentence

Every intervention is a claim about what failed. Diagnose first; choose the axis second.

1. Name the shift.
2. Choose the intervention.
3. Define the estimand.
4. Pre-declare the gate.
5. Price the decision.

The bridge

Rent the pretraining; own the corpus, target, and evaluation



Two facts from the scaling reference:

- Pretraining cost scales as $C \approx 6ND$: parameters times training tokens.
- Loss curves are *fitted response surfaces*, not laws of nature.

The allocation follows: rent pretraining; spend on the corpus, target, and evaluation a provider cannot own for you.

The Lagrangian, the isoquants, the serving bill: reference deck pp. 11–15 and 47–52, with `scaling-law-lab.html`.

The bridge

Post-training changes behaviour before you adapt anything

A downloaded checkpoint already contains two interventions:

- SFT: demonstrations teach what an acceptable reply looks like.
- Preferences: pairwise comparisons fit a latent reward (Bradley–Terry) and tilt the policy against a KL charge:

$$\pi^*(y \mid x) \;\propto\; \pi_{\text{ref}}(y \mid x)\, e^{r(x,y)/\beta}$$

π_{ref}	the SFT checkpoint — where the tilt starts
π^*	the aligned policy — the weights you download
r	the fitted reward — a *proxy* for quality

β prices the divergence. Derivations, DPO, PPO: reference deck pp. 21–37 and `colabs/rlhf_dpo_lab.ipynb`.

Why you are shown this at all

The exponential re-weights π_{ref}; it never leaves it. Every response the SFT model would have produced keeps positive probability. So adaptation starts from a *post-trained* object, and you inherit its helpfulness and its defects together.

And r is a proxy

Optimize a fitted reward hard enough and true quality falls — Goodhart. The handout's specification-search frame is the same problem with your own metric: an optimized measure stops measuring.

The bridge

Helpfulness can buy confident claims the base cannot support

When demonstrations contain facts the base model does not know:

- helpfulness rises;
- out-of-distribution truthfulness falls;
- the model learns to sound informed past what it knows.

The failed fix is the lesson

Delete the unfamiliar claims: truthfulness recovers, but helpfulness collapses. The trade-off lives in the *data*, not the optimizer.

What worked (UNIT):

- keep the response intact;
- append a reflection identifying uncertain claims;
- split uncertainty at a fixed quantile.

Helpfulness held; certain and uncertain claims separated. The agentic session turns this into abstention.

Ours: *The Helpfulness–Truthfulness Dilemma* (ACL 2025). Truthfulness scored by FactScore and WildFactScore, out of distribution.

The design space

Four axes, chosen independently

where the signal enters	prompt	retrieved evidence	weights
which objective	next-token	supervised	preference
how far weights move	frozen	LoRA, rank r	full
which base	general	domain-pretrained	proprietary

Not a ladder

Prompting, retrieval, DAPT, and SFT solve different failures. Moving weights is not “more serious” than supplying evidence.

Most useful systems are mixtures. The figure returns as each choice is settled.

I Diagnose the shift

Name which distribution moved. Different shifts admit different corrections, and one of them admits none.

The shift

Three distributions can move — and one thing that is not a distribution

Source law p_S, target law p_T, over x = the filing you read and y = the event you defined. There are two ways to write the joint,

$$p(x, y) \;=\; p(x)\,p(y \mid x) \;=\; p(y)\,p(x \mid y),$$

and a shift names which factor moved. The other is the anchor a correction leans on.

Name	Let move	Hold fixed
covariate	$p(x)$	$p(y \mid x)$
label / prior	$p(y)$	$p(x \mid y)$
concept	$p(y \mid x)$	nothing

evidence — not on this list. Here $p_S = p_T$: nothing moved except which documents you read.

One sentence, four failures

"We continue to monitor developments in climate-related regulation."

1. covariate — web prose $\rightarrow$ Item 1A
2. label — 10% exposed $\rightarrow$ 30%?
3. concept — signalled awareness $\rightarrow$ now mandatory
4. evidence — they also filed an 8-K

A new 10-K is an observation, not proof that a population law moved. Diagnose the failure, not the corpus.

1. Covariate shift

Inverse-propensity weighting — with the familiar variance problem

Row 1 of the table: let $p(x)$ move, hold $p(y \mid x)$ fixed.

Proposition 1

If $p_T(y \mid x) = p_S(y \mid x)$ and $p_T^X \ll p_S^X$, then for any predictor $h(x) \approx y$, its average loss is:

$$\boxed{R_T(h) = \mathbb{E}_{(x,y)\sim p_S}\left[w(x)\,\ell(h(x), y)\right], \quad w(x) = \frac{p_T(x)}{p_S(x)}}$$

$p_T^X \ll p_S^X$ reads: no filing in the target that the source could never have produced.

This is inverse-propensity weighting: the target risk of a *fixed* h, computed from source data. The identity is exact; finite variance is a separate assumption.

The familiar warning

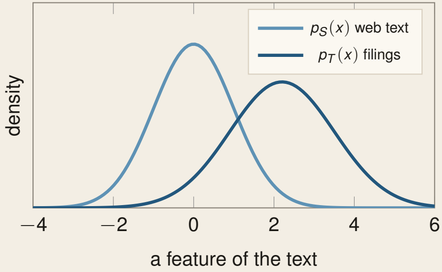
Var scales with $\mathbb{E}_S[w^2]$. If filings sit far from web text, a few documents carry the estimate and the effective sample size collapses (Sugiyama et al., 2007).

What you can actually reweight

Not the provider's pretraining: you do not have it. Your fine-tuning set, your gold sample, and — the one that matters here — your evaluation, so a source-distributed test set estimates deployment risk.

1. Covariate shift

Overlap is a precondition — and the one thing here you can actually measure



The inputs move; the labelling rule does not.

- Strong overlap: weighting may transport.
- Weak overlap: weights explode.
- No overlap: the target is extrapolation.

The cheap one

Every other correction here needs p_T / p_S, base rates, or target labels. This needs two folders and any classifier.

Label source 0, target 1; balanced split, held-out error ϵ:

$$\boxed{\hat{d}_{\mathcal{A}} = 2(1 - 2\epsilon)}$$

Accuracy rescaled into a distance: $1 - 2\epsilon$ estimates total variation, the 2 puts it on $[0, 2]$ — the units of the bound on p. 49.

Then read what separated them.

2. Label shift

The prior correction, derived — and it is one line of arithmetic

Row 2 of the table: let $p(y)$ move, hold $p(x \mid y)$ fixed.

Proposition 2 (Saerens et al., 2002)

If $p_T(x \mid y) = p_S(x \mid y)$, *then*

$$\boxed{p_T(y \mid x) \;\propto\; p_S(y \mid x)\,\frac{p_T(y)}{p_S(y)}}$$

The base rate $p(y)$ is the share of the label in a population *before* you read anything: 10% of filings risky in training, 30% in the book you are scoring. Only that ratio enters.

The base rates move *by assumption*; $p(y \mid x)$ then moves as a consequence — which is the table's point. It lists what you hold fixed, not what stayed still.

A worked number

A filing the model scores 0.50 — a coin flip. Re-weight each side by how much more common it has become:

- risky: $\frac{0.30}{0.10} = 3$, so $0.5 \times 3 = 1.5$
- safe: $\frac{0.70}{0.90} = 0.78$, so $0.5 \times 0.78 = 0.39$
- normalise: $\frac{1.5}{1.5+0.39} = 0.79$

No labels, no gradient, no retraining. Against a 0.75 review trigger, that filing just moved from ignored to reviewed.

3. Concept shift

The one that admits no correction

If $p(y \mid x)$ moved, source reweighting cannot identify the new conditional. In finance:

- a disclosure regime changes what a sentence *implies*;
- firms learn which phrasings are penalised and stop using them;
- the mapping from language to outcome is endogenous to being modelled.

The identification problem

You need target labels or an identifying assumption. There is no third source of information.

This is the Lucas critique in text: once the measure is used, behaviour adapts. Section IV shows it in climate disclosure.

The route

Axis 1 — where the signal enters

where the signal enters	prompt	retrieved evidence	weights
which objective	next-token	supervised	preference
how far weights move	frozen	LoRA, rank r	full
which base	general	domain-pretrained	proprietary

Three corrections and a purchase

- 1. Covariate $\rightarrow$ reweight, subject to overlap.
- 2. Label $\rightarrow$ correct the prior — ranking survives, the threshold count does not.
- 3. Concept $\rightarrow$ *buy* target labels, or assume identification.
- Evidence $\rightarrow$ not a shift. Supply the facts.

Rows 1–2 rearrange data you have. Row 3 and evidence need data you do not.

Still open: the objective, the update constraint, and the base model.

II The design space

Four axes, chosen independently: where the signal enters, which objective supplies it, how far weights may move, which base you start from.

The space

Prompting: change the conditioning set, for one call

A prompt changes the conditioning set for one call:

$$p(y \mid \text{instruction}, \ \text{examples}, \ x).$$

- Nothing is stored.
- Definitions and examples specify the task.
- Change the instruction and the estimand may change silently.

What prompting is good at, precisely

Specifying a task. Definitions and worked examples can identify the intended mapping; an instruction cannot create knowledge the model never saw.

If a paraphrase changes the published number, the prompt is part of the instrument. The build recipe makes that a gate.

The space

Retrieval: change the evidence — and inherit a second model

Retrieval conditions on documents fetched at query time:

$$p(y \mid x) = \sum_{z \in \mathcal{D}} p_\theta(y \mid x, z)\, p_\eta(z \mid x),$$

truncated to a top-k set and renormalized. The retriever p_η is a second model: chosen, versioned, and evaluated separately.

The right way to think about it

Retrieval primarily fixes missing evidence. It supplies facts; it does not by itself teach the model to read filings.

It updates without retraining and scales beyond one context window. The agentic session audits every stage.

The space

DAPT: same objective, different corpus

Domain-adaptive pretraining continues the base model's own objective on unlabelled in-domain text (Gururangan et al., 2020):

$$\mathcal{L} = -\sum_t \log p_\theta(x_t \mid x_{<t}), \qquad x \sim p_T,$$

or the masked-token analogue for encoders (Devlin et al., 2019). No labels are required. The intended gain is a better representation of the target register; the gain must be measured.

Three gates before DAPT

1. The failure is a register problem, not missing facts.
2. Unlabelled in-domain text is abundant.
3. A locked task suite improves out of sample.

More domain text is not automatically better. The next frame is the counterexample.

DAPT

Which corpus? We ran it — and more was worse

Our Climate-ModernBERT: one encoder, three candidate corpora, nine climate benchmarks.

- academic alone: 75.3 average F_1 (base 73.5);
- add synthetic: 74.8; add web as well: 74.1;
- commitment detection under the full mixture: 71.8 $\rightarrow$ 59.5.

Out-of-register text did not merely dilute the signal; it damaged it.

And the winner was not a corpus

Weight-averaging three independently specialized checkpoints beat joint training on the union: best overall 76.3, +2.8 over the base.

DAPT is not “pour in domain text.” Corpus choice has a sign and must be validated.

Ours: *Climate-ModernBERT* (under review, 2026).

The space

Supervised fine-tuning learns a task boundary, not a domain

SFT trains on labelled pairs:

$$\max_{\theta} \sum_{i} \log p_{\theta}(y_i \mid x_i).$$

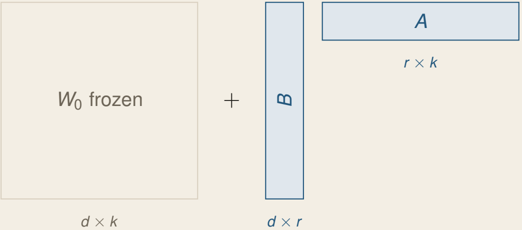
- It learns the target task boundary.
- It can address concept shift because it consumes target labels.
- Those labels are the scarce input in finance.

The trap it sets

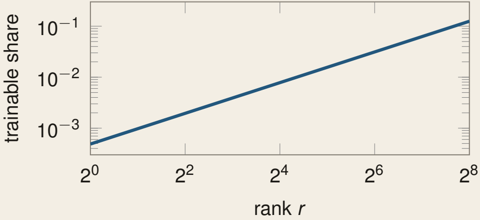
A benchmark can rise because the model learned annotator conventions. Build the evaluation independently — and before the adapter.

LoRA (Low Rank Adaptation)

The update, drawn



$W = W_0 + BA$; only the shaded blocks train.



$d = k = 4096$: $r = 8$ trains 0.39% of one matrix.

A pulls the input down to r directions
B delivers those r back to the output

The adapter is the pair. B starts at zero, so training begins exactly at W_0.

The city

W_0 is a city you may not rebuild. LoRA grants r new roads across it — A the on-ramps, B the off-ramps. Close every off-ramp and the city behaves exactly as before: that is $B = 0$.

LoRA

Rank one, and no leash at all

Prediction

The adapter is pinned at $r = 1$ — the most constrained one there is. Is there now *any* bound on how far the adapted weights can move from the base? Commit to yes or no.

Set $B(c) = cB_0$ with A fixed. For every $c \neq 0$,

$$\operatorname{rank}\left(B(c)A\right) = 1,$$

$$\|B(c)A\|_F = |c|\,\|B_0A\|_F \longrightarrow \infty.$$

One road, any length. The permit capped how many you may build — never how far they go.

Rank is a count, not a leash

r caps how many factors the update may use, never how large they get — and so never the distance from W_0.

LoRA is not a safety mechanism. If distance matters, constrain $\|BA\|$ or the output divergence explicitly.

What a rank sweep *does* diagnose — approximation capacity — is on p. 65.

The route

All four axes — and the failure picks the axis, not the reverse

where the signal enters	prompt	retrieved evidence	weights
which objective	next-token	supervised	preference
how far weights move	frozen	LoRA, rank r	full
which base	general	domain-pretrained	proprietary

What you measure	The move
change the system	
asserts a fact in no retrieved document	retrieval
holds on news, falls on filings	DAPT
matches the benchmark, not your gold sample	SFT
change only the reading	
ranks well, probabilities do not	calibration
right on average, wrong base rate	prior correction

III Finance: task before model

Define the financial decision first. The model is the last choice, not the first — and the label defines the estimand.

The estimand

Three "sentiment" models, three different objects

Label source	Estimates	Valid for
Loughran–McDonald	dictionary tone	replicating a literature
human annotation	perceived tone	communication studies
realized returns	$p(r > 0 \mid \text{text})$	trading, if it survives

All three get called "financial sentiment". They are different random variables, and a model fitted to one does not estimate another.

The foundations-session beat, in finance clothes

The head gives p over strings. The label determines which economic event the reported number estimates.

Start with the decision. Then define the label that would make the model output relevant to it.

Calibration

Temperature scaling cannot change what the ranking already knows

Proposition 3

For any $T > 0$, $\mathrm{softmax}(z/T)$ preserves the ordering of z within an example. Hence the predicted label and top-1 accuracy are invariant; NLL, Brier and ECE are not.

Proof. $z_i > z_j \iff z_i/T > z_j/T \iff e^{z_i/T} > e^{z_j/T}$; the denominator is shared and positive, so $p_i > p_j$. $\square$

Binary case only: the logit margin $z_1 - z_0$ is scaled by $1/T$, so its ordering *across* examples also survives — and ROC/AUC with it.

Do not overstate the invariance

In multiclass one-vs-rest evaluation, probabilities are compared *across examples*; temperature can reorder them. Only within-example ordering is guaranteed.

Fit T on a calibration split, never the test split (Guo et al., 2017). An exact multiclass counterexample is in the backup.

Calibration

Murphy's decomposition says exactly what temperature can buy

For binary forecasts p with outcome y, write $\bar{o}_p = \mathbb{E}[y \mid \text{forecast} = p]$ and $\bar{o} = \mathbb{E}[y]$:

$$\underbrace{\mathbb{E}(p-y)^2}_{\text{Brier}} = \underbrace{\mathbb{E}(p-\bar{o}_p)^2}_{\text{reliability}\downarrow} - \underbrace{\mathbb{E}(\bar{o}_p-\bar{o})^2}_{\text{resolution}\uparrow} + \underbrace{\bar{o}(1-\bar{o})}_{\text{uncertainty}}$$

(Murphy, 1973). Uncertainty is a property of the world; you cannot touch it.

Now the previous proposition has teeth

In the binary case temperature scaling is a *bijection* of the forecast, so the population grouping — and *resolution* — is unchanged. Any Brier improvement must therefore come through reliability.

Fitting T by NLL need not reduce the Brier reliability term in finite samples. Calibration repairs confidence; improving resolution requires new information.

Live

Which metrics move under temperature, and which cannot

Predict first, before revealing the fitted T

Classify each as invariant or changeable under $T > 0$: predicted labels, binary AUC, NLL, Brier, ECE, and the count crossing an 85% action threshold.

The action count is the trap: it depends on the probability threshold, not the ranking.

Operational consequence

Recalibration can change who receives review without changing who ranks above whom.

▷ *live lab:* `calibration-lab.html`

IV Climate: standards and strategic language

Climate disclosure is the hard case: strategic language, moving standards, and a measurement that changes the thing measured.

The domain

Climate disclosure breaks the stable-label assumption

- horizons — claims are about 2030 and 2050, so ground truth arrives after every career;
- strategic language — disclosure is voluntary in effect, and firms choose what to say;
- moving standards — TCFD, then IFRS S2 and ESRS: the taxonomy itself is a moving target (IFRS Foundation, 2024; European Commission, 2026);
- no natural label — there is no return series for "was this disclosure adequate".

Which shift is this?

All four can move — especially concept shift, because firms choose the language knowing it will be read and scored.

The calibration lab assumed a stable mapping from text to event. Climate disclosure shows why that assumption must be tested.

Weights

ClimateBERT: putting the register into the base

Our ClimateBERT continues pretraining a general encoder on climate reports, disclosures, and news (Webersinke et al., 2022).

- Objective unchanged; corpus changed.
- Intended gain: better representation of the climate register.
- Not supplied: current facts or a definition of good disclosure.

What a domain base buys

Better representations of a register general corpora contain thinly, and potentially fewer scarce labels downstream.

DAPT targets unusual language. Retrieval targets unusual or changing facts.

Measurement

Cheap talk: what a classifier makes visible

A TCFD classifier turns disclosures into a panel:

- which categories are disclosed specifically, and which only in generalities;
- whether firms with more to hide disclose more text and less substance — our cheap-talk result (Bingler et al., 2022, 2024).

The methodological move worth copying

The classifier is an instrument that produces a variable. The economic design produces the result.

Validate the instrument on held-out human-coded data before interpreting the panel.

The production version is our ChatReport: TCFD-structured analysis of sustainability reports, with the same validation discipline.

Measurement

Net-zero claims: separating a target from an aspiration

Compare:

- “We are committed to net zero.”
- “We will cut Scope 1 and 2 emissions 46% by 2030 against a 2019 baseline.”

Same sentiment; different commitment. The label schema must encode target versus aspiration, scope, baseline, and horizon (Schimanski et al., 2023).

Where schemas come from is itself our research: Co-DETECT (EMNLP 2025 demo) has the model flag the edge cases a codebook misses, and the human rewrite the codebook.

Where the value came from

Defining the estimand precisely enough to make annotation possible.

The same distinction appears in covenants, guidance, and forward-looking statements.

Evidence

ChatClimate: keep the authority in the context

Our chatClimate retrieves from IPCC reports and answers only from retrieved passages, with citations (Vaghefi et al., 2023).

- Axis: evidence, not weights.
- Corpus supplies authority.
- Citations supply auditability, not truth.
- IPCC experts scored answer accuracy.

Why this is the right default for regulated work

A retrieved answer can be traced, updated, and shown to a supervisor. A parametric-memory answer cannot.

The cost is a retriever you must own and evaluate. The agentic session audits that pipeline.

The reflexive problem

Measuring disclosure changes disclosure

Once a text measure is published, firms can optimize against it:

- more specific-sounding language;
- no corresponding increase in substance;
- x chosen strategically against a known $\hat{y}(x)$.

That is concept shift induced by the measurement itself.

The consequence for research design

A text measure has a shelf life. Re-validate it on fresh human-coded data; a stable score can conceal strategic drift.

The reflexive problem is why the next section starts with temporal splits and fresh labels.

Live instance: we are measuring ex ante what the EFRAG ESRS simplification removes, across three years of European nature disclosures — the standard itself is the moving object. Ours: working paper, 2026.

V Honest evaluation

The evaluation is the product. Leak-proof splits, calibration and abstention as first-class metrics, and AI labels treated as measurements with error.

Splits

A random split estimates the wrong quantity

Deployment is forward: use filings available at t to predict an event at $t + h$. A random split estimates instead

$$\mathbb{E}\left[\text{error} \mid \text{train and test interleaved in time}\right],$$

a different quantity. Boilerplate, persistent firm identity, and autocorrelated outcomes usually make it optimistic.

Three clocks to lock

Time: train strictly before test; gap at least the label horizon.
Entity: disjoint firms for new-firm claims; report seen firms separately.
Index: retrieval content only as of the query date.

Whether you need disjoint *firms* depends on what you are claiming. The other two clocks are unconditional.

AI labels

An LLM label is a measurement, and measurement error attenuates

Let $y \in \{0, 1\}$ be truth and $\hat{y}$ the model label, with error rates that do not depend on x — non-differential:

$$\Pr(\hat{y}{=}1 \mid y{=}0, x) = a, \qquad \Pr(\hat{y}{=}0 \mid y{=}1, x) = b.$$

Proposition 4

With $p(x) = \Pr(y{=}1 \mid x)$*,*

$$\mathbb{E}[\hat{y} \mid x] = a + (1 - a - b)\, p(x).$$

In a correctly specified linear probability model $\mathbb{E}[y \mid x] = \beta_0 + x^\top \beta$*, the slopes become* $(1{-}a{-}b)\beta$*.*

Condition on y: $\mathbb{E}[\hat{y} \mid x] = (1 - b)p(x) + a(1 - p(x))$; collect terms.

A number to remember

At $a = 0.05$, $b = 0.10$, slopes are 15% too small. More data does not remove the bias — it estimates the wrong number more precisely.

Scope: linear-probability slopes under non-differential error. No universal factor carries to logit or probit.

AI labels

A validation sample repairs the estimand — not the model

Hand-code a random subsample; estimate $\hat{a}$, $\hat{b}$; invert:

$$\boxed{\hat{p}(x) = \frac{\mathbb{E}[\hat{y} \mid x] - \hat{a}}{1 - \hat{a} - \hat{b}}}$$

Consistent when $a + b \neq 1$. The familiar orientation $a + b < 1$ just says balanced accuracy exceeds one half.

The model did not improve. The estimand was repaired (Gentzkow et al., 2019).

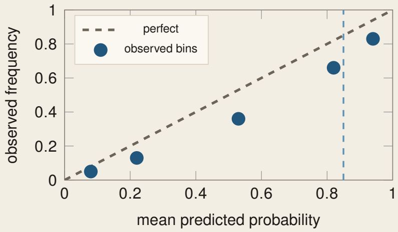
What it costs

Variance inflates by $1/(1 - a - b)^2$: at $a + b = 0.15$, a factor of 1.38. Carry the uncertainty in $\hat{a}$, $\hat{b}$ into the standard errors.

Our Dow Jones climate-discourse study validates LLM extractions against 2,000 gold-annotated articles before reading any trend.

Calibration

The reliability diagram, and why ECE is not one number



Below the diagonal: the model claims more than it delivers. Bins are unjoined and unevenly filled — there is nothing continuous to interpolate.

Expected calibration error averages the bin gaps,

$$\mathrm{ECE} = \sum_b \frac{n_b}{n} \left| \mathrm{acc}(b) - \mathrm{conf}(b) \right|.$$

Two ways it misleads

Binning: too few hides structure; too many adds noise — and nothing in the data picks the number.
Aggregation: your threshold lives in *one* region. The average can look fine while every action you take is wrong.

Report the diagram, a proper score, and the distribution of forecasts. ECE alone is not a decision metric.

Decisions

Two thresholds, both set by the loss — neither by the model

Proposition 5 (act)

With cost c_{FP} for a false positive and c_{FN} for a false negative, predicting positive is optimal iff

$$\boxed{p(y{=}1 \mid x) \;>\; \frac{c_{\mathrm{FP}}}{c_{\mathrm{FP}} + c_{\mathrm{FN}}}}$$

Proposition 6 (or decline: Chow, 1970)

If declining costs $c \in (0,1)$ against 0 correct and 1 wrong, answer with $\arg\max_y p(y \mid x)$, abstaining otherwise, when

$$\boxed{\max_y p(y \mid x) \;\geq\; 1 - c}$$

Where 0.5 comes from

Only from $c_{\mathrm{FP}} = c_{\mathrm{FN}}$. At $c_{\mathrm{FN}} = 20c_{\mathrm{FP}}$ the threshold is 0.048. Reporting accuracy at 0.5 silently assumes equal costs.

Not a risk-neutrality assumption

Replace the costs by utility differences $\ell = u(W) - u(W{-}L)$ and the rule is unchanged.

Neither rule touches the model. Both read a probability. A 0.90 cutoff on an uncalibrated score implements an unknown rule.

VI Build recipe

Build the evaluation before the adapter, price the whole system, and choose each axis from a failure you measured.

Order of work

Build the evaluation before the adapter

The previous section ended on the decision. The recipe starts there.

1. write the decision and its costs;
2. define the estimand — which event the probability is about;
3. build the split: temporal, entity-disjoint, as-of;
4. hand-code a gold sample and fix the paraphrase gate;
5. *only now* measure the base model and read the failure;
6. choose the axis that removes *that* failure;
7. re-measure, including on the general suite.

Why this order

Steps 1–4 survive every model change. Build them once and each model choice becomes a measurement. Adapt first and the benchmark moves with the model.

Before the break

Take one published text-model number. Ask questions 1–4 of it. Where does its claim first fail?

`colabs/01_calibration_changes_the_decision.ipynb` — temperature scaling and risk–coverage on real data

Cost

The system's bill, not the adapter's

Item	Frequency	Scales with
DAPT / LoRA run	once	GPU-days
serving	per query	$2N$ per token
retrieval index	continuous	corpus growth
rescore locked gold	per version	compute only
new gold coding	on drift	expert hours
re-validation	per version	expert hours
drift monitoring	continuous	people

Separate two costs

Re-score: every model version, on the locked gold set. Compute.
Re-code: only for drift, construct change, test-set wear, or new periods. People.

The two highlighted rows are the ones proposals omit — and they are the bill for "buy target labels".

The evaluation is the longest-lived asset: it outlives the models it scores.

The lender

Three linked estimators, not one model

evidence
retrieve as-of the filing date

retriever, versioned

measurement
calibrated $p(\text{exposed} \mid \text{text})$

gold-validated, de-attenuated

decision
costs, threshold, abstention

Chow's rule

Each layer has its own estimand, validation set, and failure mode.

The chain of authority

Evidence licenses the claim.
Gold labels license the probability.
Costs license the action.

Break any link and the layers below it are unlicensed — a calibrated probability over evidence that did not exist at the decision date is still a number, and still worthless.

Session III audits every stage of that chain, and turns the bottom box into a declared action or an abstention.

Carry forward

Three things from this session

1. Diagnose before you adapt. Each failure has a different correction; evidence mismatch is not distribution shift.
2. Rank constrains directions, not magnitude. LoRA is not a safety mechanism.
3. AI labels are noisy measurements. A gold sample repairs the estimand; model agreement does not.

The discipline

Calibration is frequency agreement for a specified event and population. Without labels, it is not measurable.

The agentic session applies the same discipline to evidence and authorized actions.

Next

From a measurement to a decision

The output is now a validated measurement built on point-in-time evidence. The agentic session turns it into something that *acts*:

> a time-indexed trajectory ending in a declared action or an abstention

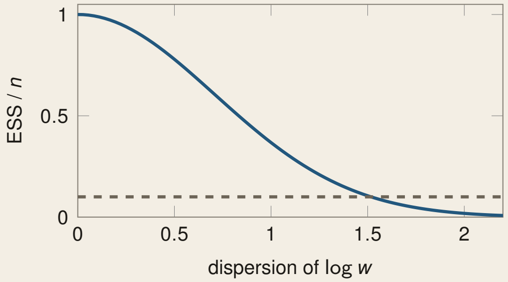
with retrieval audited stage by stage, tools under authorization, and a decision certificate.

Backup appendix

Optional detail preserved for questions and handout use — not part of the taught path.

Covariate shift

The effective sample size, and how fast it collapses



$\mathrm{ESS}/n = e^{-\sigma^2}$ for log-normal weights (verified by simulation).

Proposition 7

With weights w_i, the effective sample size is

$$\mathrm{ESS} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2},$$

and for $\log w \sim \mathcal{N}(\mu, \sigma^2)$, $\mathrm{ESS}/n \to e^{-\sigma^2}$.

The number that kills it

At $\sigma = 2$ — illustrative, not measured — 2% of your sample is doing all the work. Ten thousand documents behave like two hundred.

The bound

What theory actually promises about transfer

Proposition 8 (Ben-David et al., 2010, binary classification, 0–1 loss)

For any h in a class $\mathcal{H}$, over a representation ϕ,

$$\varepsilon_T(h) \;\le\; \varepsilon_S(h) + \tfrac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}\big(\phi_\# p_S^X, \phi_\# p_T^X\big) + \lambda,$$

$\lambda = \min_{h'} \big[\varepsilon_S(h') + \varepsilon_T(h')\big]$.

- ε_S — fit the source better;
- the divergence — change ϕ. It is the pushforward that moves: DAPT cannot alter the raw p_S^X, p_T^X, and self-supervised loss does not optimize this term;
- λ — not estimable without target labels.

Read λ carefully

The error of the *best* hypothesis on both domains at once. If no single rule does well on web text and on filings, transfer cannot work — however clever the method.

It is not estimable from unlabelled target data, and it is not a property of nature either: λ depends on $\mathcal{H}$ and on ϕ, so changing the representation moves the obstruction as well as the divergence.

LoRA

Restrict the update to a low-rank subspace

Freeze $W_0 \in \mathbb{R}^{d \times k}$ and learn a factored update (Hu et al., 2021):

$$\boxed{W = W_0 + \frac{\alpha}{r} BA, \qquad B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k}}$$

with B initialized to zero, so training starts exactly at the base model.

Proposition 9

The update has $r(d+k)$ parameters against dk. At $d = k = 4096$ and $r = 8$ that is 65,536 against 16.8M — 0.39%.

Two properties that matter operationally

BA can be merged into W_0 after training, so inference latency is unchanged. And many adapters can share one base in memory — one model, many domains.

The α / r factor is the *original* convention. Treat α, r and the learning rate as coupled: at high rank $1/r$ can produce gradient collapse, which is why rank-stabilized variants scale by $\alpha / \sqrt{r}$ instead (Kalajdzievski, 2023).

LoRA

Why a low rank suffices at all

The empirical motivation is intrinsic dimension: fine-tuning can be done in a random low-dimensional subspace of *parameter space*, and the dimension needed falls as the base model grows (Aghajanyan et al., 2021).

Note the gap in that argument. A low-dimensional *parameter* subspace does not imply that each weight matrix's update is low rank — those are different objects. Suggestive motivation, not a derivation.

The honest status of that claim

An empirical regularity with a plausible story, measured on specific task suites. Not a theorem, and no way to know the right r for *your* task except by sweeping it.

Note also that $\{BA : \mathrm{rank}(BA) \leq r\}$ is not a fixed subspace — it is a non-linear variety, and LoRA learns which row and column spaces to use. That is why it works better than freezing a random projection.

Cost

QLoRA, and where the memory actually goes

Full fine-tuning of an N-parameter model needs roughly $16N$ bytes in the standard mixed-precision recipe: fp16 weights and gradients ($2N+2N$), two fp32 Adam moments ($8N$), and the fp32 master copy ($4N$) — implementation-dependent in detail. For $N = 7$B that is ≈ 112 GB before activations.

QLoRA quantizes the frozen base to 4 bits and trains only the adapter in higher precision (Dettmers et al., 2023): base $\gtrsim 0.5N$ bytes — more, once quantization metadata is counted — and optimizer state only on $r(d+k)$ parameters. Activations and temporary buffers remain.

The reframing

Optimizer state, not weights, is what makes fine-tuning expensive. LoRA removes it for 99.6% of the parameters; quantization then shrinks what remains.

Result: a 7B model adapts on one consumer GPU. That is the whole reason domain adaptation is available to a university group at all.

Cost

Adaptation is not free: what you lose

Moving weights toward a domain can *interfere* with what they already encoded — not inevitably, and not uniformly. Measured symptoms:

- forgetting — general reasoning and instruction-following degrade;
- calibration drift — the probabilities shift even where the argmax does not;
- narrowing — the model becomes confident on the fine-tuning distribution and worse at flagging what lies outside it.

The measurement rule

Evaluate the adapted model on the *general* suite too, not only the domain one. A domain gain of 4 points bought with a 9-point general loss is a bad trade you will not otherwise see.

This is why the retrieval axis is often the right first move: it changes no weights, so it cannot forget.

FinBERT

An overloaded name, not one recipe

At least three published things are called FinBERT, and they differ on *two* axes:

Name	**Base**	**Domain stage**	**Task stage**
Araci	BERT	continued pretraining on TRC2	PhraseBank, FiQA
Yang et al. BaseVocab	BERT	4.9B financial tokens	finance tasks
Yang et al. FinVocab	random init	financial pre-training from scratch	finance tasks

So “we used FinBERT” does not identify a model (Araci, 2019; Huang et al., 2023).

What to write in a paper

The checkpoint hash, the label source, the split rule, and the date. Without those four, the result is not reproducible even by you, six months later.

This is not pedantry: the three variants disagree materially on the same sentences, and which one you used is a bigger effect than most of the treatments people test.

Economics

BloombergGPT is a case about proprietary data, not architecture

A 50B model trained on roughly half financial and half general tokens — 363B against 345B (Wu et al., 2023). Those counts are the paper's; that the architecture is unremarkable and the corpus unbuyable is *our reading*, not a result.

Read it through the reference deck's transfer law: the value of a private corpus is the effective data it transfers that a public corpus cannot supply — and that is a quantity you can estimate before spending anything.

The counterfactual nobody ran

Would DAPT plus LoRA on an open base, at a fraction of the cost, have closed most of the gap? The paper cannot answer it, and neither can we — which is itself the finding.

For a university group the practical lesson is the reverse of the headline: the axis with the best return is usually evidence, not weights.

Contamination

Two different problems that get one name

Backward contamination — the test set was in pretraining. The score mixes memorization with ability, and no post-hoc analysis separates them.

Forward leakage — the model saw *the future* relative to the prediction date: a later filing, a later news story, an index built today.

Only one has a clean defence

Backward: use data created after the model's cutoff. Forward: reconstruct the information set as of the decision date, everywhere — including the retriever.

A model card's cutoff date is therefore a research input, not a footnote. If you cannot state it, you cannot claim either defence.

AI labels

Agreement is not identification

A common defence: "two models agree on 94% of cases." It identifies nothing.

Shared training data, prompt style and tokenizer make correlated errors plausible — but that is not the whole objection, because even *conditionally independent* classifiers cannot have both their accuracies recovered from a single agreement rate. One equation, two unknowns.

What does identify

A gold sample: human-coded, randomly drawn from the deployment distribution, coded by people who did not see the model's output, with an inter-coder agreement statistic reported.

Agreement is a consistency check under *either* dependence or unknown independent accuracies. Consistency checks catch bugs; they do not license inference.

Robustness

Paraphrase stability is a gate you can fail

If the instrument is a frozen prompt, then the prompt is part of the instrument. Test it: generate k semantically equivalent phrasings and re-run the whole pipeline.

Report the dispersion of the estimate across phrasings alongside the point estimate. If it exceeds your effect size, you have measured the prompt.

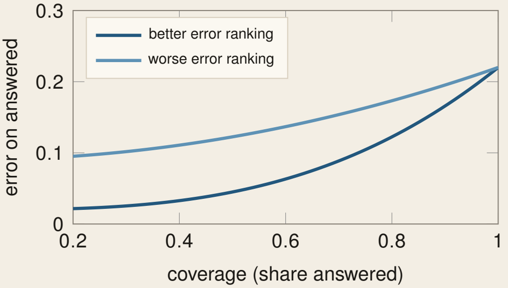
Pre-declare the gate

Fix the paraphrase set and the tolerance *before* looking at results, exactly as you would pre-register a specification.

The published evidence on how often this gate fails is thin — one unreplicated single-author study (Dixon, 2026). Adopt the discipline; do not cite an effect size.

Abstention

Report a risk–coverage curve, not an accuracy



Stylized. Two models with the *same* full-coverage error.

At full coverage both models look identical. Only when they may decline does the difference appear — one ranks its errors better.

Calibration is a different property

The curve depends only on the score's *ordering*, so binary temperature scaling leaves it exactly unchanged. A constant base-rate forecast is perfectly calibrated with no selective value; a monotonically distorted score can be badly calibrated and rank errors perfectly.

Publish the curve and your cost-implied operating point — not a bare AURC, which also absorbs the full-coverage error.

Gold samples

How many cases must be hand-coded

The correction needs $\hat{a}$, $\hat{b}$ with usable precision. Each is a binomial proportion *within its own class*, so

$$n_0 \approx \frac{a(1-a)}{\mathrm{SE}^2}, \qquad n_1 \approx \frac{b(1-b)}{\mathrm{SE}^2},$$

where n_0 counts gold cases with $y = 0$ and n_1 those with $y = 1$ — not the total.

error rate	target SE	*in-class* cases
$a = 0.05$	0.02	119
$a = 0.05$	0.01	475
$a = 0.20$	0.02	400

So 119 means 119 *true negatives*; if positives are rare, $\hat{b}$ needs a far larger draw, or stratification.

The budget line nobody writes down

A few hundred documents, coded by two people, with disagreements adjudicated: days of expert time.

The $1/(1-a-b)^2$ variance inflation assumes a, b *known*; estimating them adds delta-method or bootstrap uncertainty, and the rates must be stable between the gold sample and deployment.

Gold samples

Report an agreement statistic, and pre-declare the floor

Two coders will disagree. Krippendorff's α corrects observed disagreement for what chance would produce (Krippendorff, 2004):

$$\alpha = 1 - \frac{D_o}{D_e}.$$

The conventional cutoffs — ≥ 0.80 comfortable, < 0.667 tentative at best — are content-analysis heuristics, not inference thresholds. The floor you pre-declare should come from your decision's cost.

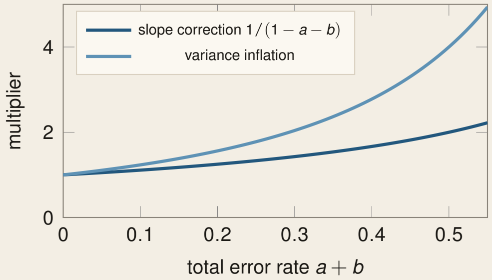
What low agreement can mean

Often an underspecified construct — but also thin coder training, class imbalance, low prevalence or ambiguous instructions. Compute α *before* adjudication; the adjudicated labels then become the gold set.

And it is diagnostic for the model too: if humans cannot agree what a "net-zero target" is, a classifier's 0.92 accuracy against one coder's labels is measuring that coder.

Attenuation

What the correction does to a coefficient, drawn



At $a+b=0.15$: slope $\times 1.18$, variance $\times 1.38$.

The correction is not free. It rescales the point estimate and inflates its variance by the *square* of the same factor.

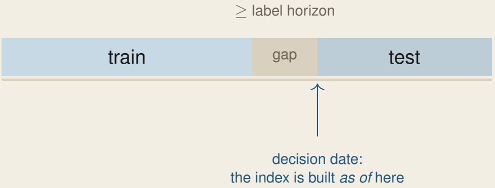
The honest reporting

Publish the uncorrected estimate, the corrected one, $\hat{a}$ and $\hat{b}$ with their standard errors, and the gold sample size. A reader can then reconstruct everything.

There is no universal cutoff: at $a+b=0.4$ variance inflates $\times 2.8$, at 0.5, $\times 4$. Whether that is tolerable depends on the sample size, the effect, and the cost of a wrong call — set the tolerance before seeing the estimates.

Splits

The timeline, drawn — and where the gap goes



The gap must be at least the label horizon, or a training example's outcome is already visible inside the test window.

The retrieval trap

Even with a perfect temporal split, an index built *today* lets the model retrieve documents that did not exist at the decision date. The split was clean; the evidence was not.

The agentic session opens on exactly this failure.

Benchmarks

They diagnose; they do not certify

A public benchmark is a sample of questions, chosen by someone else, for a purpose that is not yours, possibly inside the pretraining corpus.

It is genuinely useful for triage — ruling models out, catching regressions. It is not evidence that a system is fit for a decision, and no aggregate score can be.

We measured this (Usmanova et al., 2026): the best fact-checking system on SciFact (macro-F_1 0.700) falls to 0.315 on ClimateCheck; the 2025 shared-task winner and runner-up *swap ranks* with the metric; and TF–IDF with logistic regression beats every zero-shot LLM on the most informal corpus by 20+ points.

What supplies evidence instead

Private cases from your own deployment distribution, split temporally, scored with proper scoring rules, with a coverage curve and a validated label sample. That is evidence of fitness — “certification” is a word with regulatory meaning no internal benchmark can supply.

This is expensive, recurring, and denominated in people — the two rows of the reference deck’s cost table that projects forget to budget.

LoRA

Which matrices, and the rank sweep as a diagnosis

Not every weight matrix needs an adapter. Adapting only the attention projections W_Q, W_V often recovers most of the gain at half the adapter parameters; adding the feed-forward matrices sometimes helps.

Resist the tempting reading. Placement and rank do not identify task shift against domain shift: both also move with the optimizer, the scaling convention, data size, regularization, architecture and evaluation noise.

What the sweep does tell you

Adapter-capacity sensitivity under this training setup — and on validation, not training loss, which improves with capacity almost by construction.

That is worth two hours. It is not a diagnosis of which axis you should have been on; only a held-out comparison against DAPT answers that.

Choosing

Retrieve, adapt, or both — from the reference deck's transfer law

The effective-data *multiplier* $D_T \,/\, D_f$ declines as your own data grows ($D_T = k\, D_f^{\alpha_t} N^{\beta_t}$, $\alpha_t < 1$); the total does not literally saturate. Comparing the two axes at a fixed budget:

	retrieval	**adaptation**
fixes	missing facts	missing register
updates	instantly	per retrain
forgets	never	yes
cost	per query	once, then serve
audit	citations	weights only

The default, for this audience

Retrieval first. It is the only axis whose output is traceable to a source, and in regulated work traceability is not a nice-to-have.

Adapt when the diagnosis says register, not facts — and when you have the unlabelled corpus that DAPT needs.

Scoring

Accuracy cannot identify a probability

Proposition 10

Accuracy depends on the forecast only through $\arg\max_y p(y \mid x)$. *Hence it is not strictly proper: every forecast with the correct argmax attains the same expected accuracy.*

Proof. The 0–1 loss of a forecast is $\mathbf{1}[\arg\max p \neq y]$, which is invariant to any change of p preserving the argmax. □

So a model reporting 0.51 and one reporting 0.999 score identically — and one of them is lying.

Which is the whole argument for proper scores

Log loss and Brier are strictly proper: they are uniquely minimized at the true conditional. Accuracy, F1 and AUC are not, and none of them can be used to select a model whose probabilities you intend to act on.

The foundations deck proved the log score proper. This is that proposition, cashed.

Scoring

Sharpness subject to calibration

Murphy's decomposition suggests the operational goal: among calibrated forecasts, prefer the sharpest — the one whose predictions are most spread out.

Calibration alone is trivially achievable (predict the base rate every time). Sharpness alone is achievable by a confident liar. Only the pair is meaningful.

The reporting triple

A reliability diagram, a proper score, and the distribution of predicted probabilities. Any one alone is easy to game; together they are much harder — though test reuse and subgroup miscalibration still hide from all three.

A histogram of forecasts piled at 0.5 is a model that has learned nothing and is being honest about it — which is worth knowing before deployment, not after.

Selective risk

One number for the whole coverage curve

Let $g(x) \in \{0, 1\}$ be the decision to answer. Selective risk and coverage are

$$R(f, g) = \frac{\mathbb{E}[\ell(f(x), y)\, g(x)]}{\mathbb{E}[g(x)]}, \qquad \phi(g) = \mathbb{E}[g(x)].$$

Sweeping the threshold traces a curve; its area (AURC) summarizes the model's ability to know what it does not know.

Why the curve, not an accuracy

It scores the thing you deploy: a system that answers some cases and defers others. Full-coverage accuracy scores a system nobody should build.

Report the curve with risk at your chosen coverage and the expected decision cost — both are more decision-relevant than a bare AURC, which also absorbs the full-coverage error. The ordering it needs is a *ranking*, so it survives temperature scaling; the threshold does not.

Specification search

The garden of forking paths, in prompt space

A text pipeline has many defensible choices: the prompt, the model version, the chunking, the label schema, the threshold, the window. Each is a researcher degree of freedom, and the space is far larger than in a conventional specification search.

Reporting the best is not a result. It is a maximum of a noisy surface.

What to do instead

Pre-declare the pipeline; report a specification curve over the defensible variants; and hold out a final test set that is scored once.

This is the paraphrase gate generalized to every choice, and it is the discipline that decides whether a text-as-data finding replicates.

Backup

Vocabulary mismatch: why the tokenizer matters too

A general BPE vocabulary fragments domain terms — tickers, standard names, regulation numbers — into many tokens, which lengthens sequences, raises cost, and spreads a single concept across positions.

Extending the vocabulary requires new embedding rows, which start untrained. It pays only with enough domain data to train them, and it breaks comparability of loss with the base model.

Backup

From-scratch domain pretraining: when the economics work

From the reference deck: training is $6ND$ once and serving is $2N$ forever. A from-scratch domain model is justified when the corpus is genuinely private, large, and the served volume is high enough to amortize a smaller model.

Three conditions, all measurable in advance. In practice most institutions fail the second, and the honest answer is DAPT plus retrieval on an open base.

Calibration alone does not make a forecast informative

The constant forecast $p \equiv \bar{o}$ is perfectly calibrated and useless: in Murphy's decomposition its reliability is zero and so is its resolution.

So report both terms, or a proper score that combines them. A calibration plot alone can make an uninformative model look excellent.

Exact multiclass rank counterexample

For class 1, compare logits $A = (2, 0, 0)$ and $B = (1.5, 1.4, -10)$.
$T = 1$: $p_A = .787 > p_B = .525$.
$T = 10$: $p_A = .379 < p_B = .434$.
Each example keeps its argmax; the across-example ranking flips.

Backup

Greenwashing signals are triage, not verdicts

A model can flag disclosures whose language is vague relative to peers. It cannot establish intent, and the label “greenwashing” carries a legal meaning the classifier does not estimate.

Report the measured construct — specificity, comparability, presence of a baseline year — and let the reader supply the interpretation. This is the estimand discipline applied to naming.

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